

RONIT AGRAWAL

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EDUCATION

Bennett University

B.Tech in Computer Science and Engineering

Greater Noida, UP

2024 – 2028

Achievement: Smart BU Hackathon — Ranked **90th out of 400+** teams

PROJECTS

Cravora — Campus Food & Parcel Delivery Platform

[GitHub](#) | [Live](#)

Tech: React.js, Vite, Node.js, Express.js, MongoDB Atlas, Mongoose, JWT, Cloudinary, Render, Vercel

- Built a **full-stack peer-to-peer delivery platform** with **3 user roles** (Student, Outlet Owner, Runner), **15+ REST API endpoints**, and real-time order tracking via polling
- Secured authentication using **httpOnly JWT cookies** with bcrypt password hashing, role-based middleware, and **OTP-based password reset** with 5-minute expiry via SMTP email
- Engineered a **6-state finite state machine** for order lifecycle (**Pending** → **Accepted** → **Preparing** → **OutForDelivery** → **Delivered** | **Cancelled**) with server-side transition validation
- Designed **4 MongoDB schemas** with strategic denormalization — embedded menu subdocuments for atomic reads, point-in-time order snapshots, and Mongoose population for relational queries
- Deployed frontend on **Vercel** and backend on **Render** with cross-origin cookie support, **MongoDB Atlas** for cloud database, Cloudinary image uploads, and production-grade **CORS configuration**

Ledgerly — Expense & Finance Tracker

[GitHub](#)

Tech: React.js, Node.js, Express.js, MongoDB, Chart.js, JWT, REST APIs

- Built a **MERN-based** financial management system with real-time expense tracking and **interactive analytics dashboard** using Chart.js for spending pattern visualization
- Optimized data aggregation performance using **MongoDB Aggregation Pipelines** for computing category-wise spending summaries and monthly trend analysis
- Implemented secure authentication with **JWT tokens stored in httpOnly cookies**, input validation, and protected API routes with middleware-based access control

Smart Sales — Customer Intelligence System

[GitHub](#)

Tech: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

- Designed an **end-to-end ML preprocessing pipeline** with outlier detection, feature scaling, and encoding, reducing raw feature noise by **40%** before model training
- Applied **PCA** for dimensionality reduction, retaining **95% variance** while reducing feature space, and performed **K-Means clustering** with Elbow + Silhouette validation for customer segmentation
- Built a **Logistic Regression classifier** with hyperparameter tuning via **RandomizedSearchCV (5-fold CV)**, achieving robust generalization on unseen test data

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL

Frontend: React.js, Vite, HTML5, CSS3, Bootstrap

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases: MongoDB Atlas, MongoDB Compass, Mongoose, Aggregation Pipelines, SQL

AI/ML: Scikit-learn, Pandas, NumPy, PCA, Feature Engineering, Supervised & Unsupervised Learning

Tools: Git, GitHub, Postman, VS Code, Vercel, Render, Cloudinary, npm